

Zubair Mughal

LIVE CODING CLASSES FOR KIDS, TEENS & BEGINNERS

PY

COURSE 01 OF 12
CURRICULUM GUIDE

Python for Kids & Teens

Programming foundations & computational thinking

GRADE BAND

Grade 8–12

DIFFICULTY

Beginner

DURATION

14–16 weeks · 2 sessions/week

ABOUT THIS COURSE

Course Overview

Python is the most widely used beginner language in the world — and the same language behind AI, data science and automation. This course builds real computational thinking and programming fluency from nothing, one working project at a time.

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Prerequisites: None — this is where most students start.

LEARNING OUTCOMES

What Students Will Learn

- Read and write Python programs using correct syntax and structure
- Apply variables, data types and operators to solve problems
- Use conditionals and loops to control program logic
- Write and call functions to organize and reuse code
- Work with lists, strings and dictionaries to manage data
- Debug programs systematically using error messages and print-tracing

WHAT WE'LL USE

Tools & Technologies

Python 3 VS Code Thonny / Replit Git (introduced)

CURRICULUM

Module-by-Module Breakdown

MODULE	FOCUS
01 · Computational Thinking & Setup	Breaking problems into steps; installing Python; first program with print() and input().
02 · Variables, Data Types & Operators	Numbers, strings, booleans; arithmetic and comparison operators.

MODULE	FOCUS
03 · Control Flow	if / elif / else, boolean logic, nested conditionals.
04 · Loops & Iteration	for and while loops, loop control, simple simulations and patterns.
05 · Functions	Defining and calling functions, parameters, return values, scope.
06 · Data Structures I — Lists	Indexing, list methods, iterating over collections.
07 · Data Structures II — Dictionaries & Strings	Key-value data, string methods, formatted output.
08 · Debugging & Mini-Capstone	Reading tracebacks, testing, building and presenting a final project.

LEARNING BY BUILDING

Projects Students Will Build

- Number Guessing Game — Randomization, loops and conditional logic.
- Simple Calculator / Unit Converter — Functions and operator practice.
- Text-Based Adventure Game — Control flow, functions and lists working together.
- Quiz / Trivia App — Dictionaries, scoring logic and loops.

BEYOND THE SYNTAX

Skills Developed

Computational thinking Syntax fluency Debugging Logical reasoning Functions & abstraction

PORTFOLIO OUTCOME

Four working Python programs in a personal, GitHub-ready folder, each with a short written explanation of how it works.

RECOMMENDED NEXT STEP

JavaScript for Kids & Teens, or Web Development for Kids & Teens